

HR Database Normalization

customer_id	customer_name	phone_number	plan_name	plan_price	services	devices	tower_id	tower_location
C001	Aung Aung	09123456789	Premium	30000	Voice, SMS, Data	iPhone, Router	T01	Yangon
C002	Su Su	09987654321	Basic	10000	Voice, SMS	Samsung	T02	Mandalay
C003	Kyaw Kyaw	09777777777	Premium	30000	Voice, SMS, Data	Router	T01	Yangon

Q1. What normalization problem exists in the original HR table?

The original HR table has several normalization problems:

Problems:

- Repeating groups / multivalued attributes
 - Skills: Python, SQL
 - Certifications: AWS, Azure
- Data redundancy
 - Department and manager repeated for multiple employees
- Update anomalies
 - If manager changes, multiple rows need updates
- Insertion anomalies
 - Cannot add a department without employee data
- Deletion anomalies
 - Deleting an employee may remove department/project info

Main issue:

The table is not in First Normal Form (1NF) because attributes contain multiple values.

Q2. Convert the HR table into 1NF

emp_id	emp_name	department	manager	skill	certification	project	project_location
E01	Aung	IT	U Hla	Pytho	AWS	HRIS	Yangon

emp_id	emp_name	department	manager	skill	certification	project	project_location
	Aung			n			
E01	Aung Aung	IT	U Hla	Python	Azure	HRIS	Yangon
E01	Aung Aung	IT	U Hla	SQL	AWS	HRIS	Yangon
E01	Aung Aung	IT	U Hla	SQL	Azure	HRIS	Yangon
E02	Su Su	Finance	Daw May	Excel	CPA	Payroll	Mandalay
E02	Su Su	Finance	Daw May	Power BI	CPA	Payroll	Mandalay
E03	Kyaw Kyaw	IT	U Hla	Java	AWS	HRIS	Yangon
E03	Kyaw Kyaw	IT	U Hla	Python	AWS	HRIS	Yangon

Q3. What is the main issue after converting to 1NF?

Main issue:

Partial dependency

- Composite key may be: (emp_id, skill, certification)
- Employee name depends only on emp_id
- Department depends only on emp_id
- Project depends only on emp_id

Result: The table is not in Second Normal Form (2NF).

Q4. Convert the 1NF table into 2NF

Employee table

emp_id	emp_name	department	manager	project
E01	Aung Aung	IT	U Hla	HRIS
E02	Su Su	Finance	Daw May	Payroll
E03	Kyaw Kyaw	IT	U Hla	HRIS

Employee skills table

emp_id	skill
E01	Python
E01	SQL

emp_id	skill
E02	Excel
E02	Power BI
E03	Java
E03	Python

Employee certifications table

emp_id	certification
E01	AWS
E01	Azure
E02	CPA
E03	AWS

Q5. What issue still exists after 2NF?

Main issue: Transitive dependency

Examples:

- department $\rightarrow$ manager
- project $\rightarrow$ project_location

Problem: Non-key attributes depend on other non-key attributes.

Q6. Convert the 2NF design into 3NF

Employee table

emp_id	emp_name	department_id
E01	Aung Aung	D01
E02	Su Su	D02
E03	Kyaw Kyaw	D01

Department table

department_id	department_name	manager
D01	IT	U Hla
D02	Finance	Daw May

Project table

project_id	project_name	project_location
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project_id	project_name	project_location
P01	HRIS	Yangon
P02	Payroll	Mandalay

Employee_project table

emp_id	project_id
E01	P01
E02	P02
E03	P01

Employee_skill table

emp_id	skill
E01	Python
E01	SQL
E02	Excel
E02	Power BI
E03	Java
E03	Python

Employee_certification table

emp_id	certification
E01	AWS
E01	Azure
E02	CPA
E03	AWS

Q7. Is the HR database now in BCNF?

Yes.

Because:

- Every determinant is a candidate key
- department_id determines department data
- project_id determines project data
- emp_id determines employee data

Conclusion: The design satisfies Boyce-Codd Normal Form (BCNF).

Q8. What 4NF issue can exist in HR data?

Issue: Multivalued dependencies

Example:

- Employee can have multiple skills
- Employee can have multiple certifications

These are independent of each other.

Solution:

Separate tables:

- Employee_Skill
- Employee_Certification

Q9. What is a possible 5NF issue in HR data?

Issue: Join dependency

Example:

If:

- Employees
- Projects
- Certifications

are stored together, reconstructing valid combinations may create false records.

Solution: Use separate relationship tables and only join when necessary.

Q10. What is the final normalized HR database design?

Final Design:

Employee

- emp_id (PK)
- emp_name
- department_id (FK)

Department

- department_id (PK)
- department_name

- manager

Project

- project_id (PK)
- project_name
- project_location

Employee_Project

- emp_id (FK)
- project_id (FK)

Employee_Skill

- emp_id (FK)
- skill

Employee_Certification

- emp_id (FK)
- certification